NGUYEN THAI SON

DevOps Engineer

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Summary

I have practical experience in deploying and overseeing cloud-based, on-premise applications and services. Skilled in Python and Bash scripting, I have a strong grasp of networking and security concepts. My expertise includes cloud computing technologies, focusing on AWS, Azure, and Kubernetes (K8S) for container orchestration.

Skills

- **Programming:** Python, Bash
- **Operating Systems:** Linux, Windows
- **Relational Databases:** MySQL, PostgreSQL
- **Infrastructure as Code:** Terraform, OpenTofu, Terragrunt, Ansible
- **Cloud Services:** AWS, Azure, GCP
- **Version Control:** Git
- **Virtualization:** VMware Workstation, VirtualBox, QEMU
- **Local VM Tooling:** Lima, Vagrant (VirtualBox provider)
- **Containerization:** Docker
- **Container Orchestration:** Kubernetes (K8S, K3S)
- **CI/CD Tools:** GitHub Actions, GitLab Actions, ArgoCD, CloudBuild, Bitbucket Pipelines
- **Monitoring & Analytics:** Prometheus, Grafana, Loki, PostHog, Logstash, Elasticsearch, Kibana
- **Service Mesh:** Istio
- **Message Queue:** Kafka
- **VPN:** OpenVPN, WireGuard, Tailscale
- **Identity & Access Management (IAM / SSO):** ADFS, OIDC, SAML (Auth0, Authelia, Authentik)
- **AI Tools:** Cursor, MCP (Model Context Protocol), Cursor agents, Hindsight
- **Security & Code Quality:** Gitleaks, Semgrep, Trivy, DefectDojo, SonarQube (secrets, SAST, SBOM, container scans, vulnerability management); ESLint, Prettier (linting, formatting, pre-commit hooks)

Experience

H3Tech, Inc. *Ho Chi Minh City, 06/2025 - present*
DevOps Engineer

- **Infrastructure as Code (IaC):** Architected and deployed scalable cloud infrastructure on **GCP** using **Terraform**, including Cloud Run, Cloud SQL, Load Balancer, Cloud Armor, and Artifact Registry.
- **Secure Access Management:** Implemented a secure **Bastion Host** solution to manage private connections to Cloud SQL via **Identity-Aware Proxy (IAP)**, ensuring zero-exposure of database ports to the public internet.
- **DevSecOps Integration:** Integrated automated security scanning tools (**Trivy, Semgrep, Gitleaks**) into the CI/CD pipeline (**Cloud Build**) to detect vulnerabilities and secret leaks early in the development lifecycle.
- **Automated Monitoring & Alerting:** Engineered a robust notification system using **Google Chat** integration to provide real-time updates on build/deployment status and system health.
- **SRE & Observability:** Configured advanced monitoring for **SLO violations**, error rate spikes (HTTP 5xx $> 5\%$), and critical application logs via **Cloud Logging**, enabling rapid incident response.
- **AWS Platform Modernization:** Migrated customer infrastructure from legacy **CDKTF** to **OpenTofu** with **Terragrunt** wrappers; moved source control from **AWS CodeCommit** to **GitHub** and CI/CD from **CodePipeline** to **GitHub Actions**, with reusable workflow configs across multiple repositories.
- **Tenant Provisioning Automation:** Automated end-to-end AWS onboarding (Azure–AWS SSO, **Elastic Beanstalk**, **Lambda**, health checks, **RDS** access via **Bastion + SSM**), cutting new-tenant setup from 2 days to 2 hours; encrypted Git-managed secrets with **SOPS** via the Terraform provider.
- **Centralized CI/CD & DevSecOps Platform:** Architected a centralized, versioned CI/CD template repository in **GitLab CI** inherited across multiple project repositories, standardizing build, container push, security scans (**Trivy SBOM** and container vulnerability scans), reporting into **DefectDojo**, and automated SSH deployments for legacy VM services migrating to Kubernetes.
- **Self-Hosted Platform (K3s & ArgoCD):** Hosted internal services on a self-managed **K3s** cluster with GitOps-driven continuous delivery via **ArgoCD**, automating manifest synchronization and deployment workflows.
- **Centralized Identity & SSO:** Integrated Single Sign-On (**SSO**) across internal services (**GitLab, ArgoCD, Homarr**, and extensible to other tools) via **ADFS** and standard identity providers (**Auth0, Authelia, Authentik**).
- **Developer Hub & Knowledge Base:** Self-hosted **GitLab** for internal source code management and **GitLab Pages** for team technical documentation secured with GitLab authorization; deployed **Homarr** as a centralized service portal dashboard to streamline navigation and team onboarding.
- **Product Analytics & Observability:** Self-hosted **PostHog** to ingest user activity events for product research, track client-side errors, and replay user sessions for rapid debugging.
- **AI Agent Long-Term Memory:** Self-hosted **Hindsight** to provide persistent, long-term memory for developer and Business Analyst (BA) AI agents, preserving continuous context across sessions without repetitive knowledge re-feeding.

LTV SOFTWARE COMPANY LIMITED *Ho Chi Minh City, 06/2024 - 05/2025*
DevOps Engineer

- **Secure Networking:** Implemented **Tailscale VPN** solutions for vessel-to-shore connectivity, ensuring encrypted and reliable communication for maritime operations.
- **Service Mesh & Traffic Management:** Architected **Istio Service Mesh** on **AWS EKS** to streamline ingress traffic, mutual TLS (mTLS), and advanced routing policies.

- **CI/CD Optimization:** Engineered standardized pipelines for multi-platform repositories using **GitHub Actions** and **GitLab CI**, reducing deployment lead time.
- **Edge Computing (K3s):** Orchestrated remote service deployments and updates across vessel-based **K3s clusters**, ensuring high availability in low-bandwidth environments.
- **Observability:** Developed a comprehensive monitoring stack using **JMX Exporter, Prometheus**, and **Grafana** to visualize Java service metrics for both edge and cloud environments.
- **Centralized Logging:** Established a unified logging architecture using **Grafana Loki**, aggregating logs from all shore-based services for rapid troubleshooting.

GALAXY FINX JOINT STOCK COMPANY *Ho Chi Minh City, 10/2022 - 11/2023*
Cloud Engineer

- **Multi-Cloud IaC:** Developed reusable **Terraform** modules for **AWS** (VPC, ALB, ASG) and **Azure** (vNet, AKS), ensuring infrastructure consistency across providers.
- **Infrastructure Orchestration:** Managed complex cloud states using **Terragrunt**, utilizing **AWS S3** for backend storage and **DynamoDB** for state locking.
- **Configuration Management:** Authored **Ansible** playbooks for automated **Couchbase** cluster provisioning, user management, and **OpenVPN** setup.
- **GitOps Deployment:** Implemented continuous deployment (CD) workflows using **ArgoCD** to synchronize Kubernetes manifests with Git repositories.
- **Containerization:** Leveraged **GitHub Actions** to automate Docker image builds and security tagging for seamless integration into the container registry.
- **Monitoring:** Deployed and tuned **Prometheus** and **Grafana** dashboards to monitor system health and application performance metrics.

Education

FPT University, Ho Chi Minh City *09/2020 - 05/2024*
Major: Information Assurance
Final GPA: 8.1/10
Achievements: Good Student Award in Spring, Summer, and Fall semesters.

Passions

I am eager to join a company that invests in employee growth with structured career paths and continuous learning opportunities such as training, workshops, and certifications. I am particularly passionate about optimizing CI/CD pipelines to eliminate bottlenecks and improve efficiency.